

A continuous Tauberian theorem for Karcher means and a sharp obstruction to the unit-lag formulation

Candidate full answer for arXiv:1906.06988

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Status: candidate full answer, likely valid. The curve analogue of the source’s discrete Tauberian theorem holds in every Hadamard space under the natural metric-derivative condition. An explicit smooth real-valued curve shows that the superficially literal unit-time-lag condition is insufficient.

1 Source question

The source is Hadi Khatibzadeh and Hadi Pouladi, *Tauberian Conditions for Almost Convergence in a Geodesic Metric Space*, arXiv:1906.06988 [1]. For a sequence (x_n) in a Hadamard space, their Theorem 3.5 says that convergence of the Karcher means σ_n to y , together with

$$n d(x_n, x_{n-1}) \longrightarrow 0,$$

implies $x_n \rightarrow y$. Remark 3.1, PDF page 14, defines the Karcher mean σ_T of a curve $c : [0, \infty) \rightarrow \mathcal{H}$ as the unique minimizer of

$$\mathcal{G}_T(z) := \frac{1}{T} \int_0^T d^2(c(t), z) dt$$

and states that “Theorem 3.5 for curves remains an open question.”

The results of this section and Remark 2.1 propose this question: Does the Tauberian conditions hold for the Karcher mean in more general geodesic spaces for example uniformly convex geodesic metric spaces or in uniformly convex Banach spaces?

Remark 3.1. Let $(\mathcal{H}, d)$ be a Hadamard space. For a curve $c : [0, \infty) \rightarrow \mathcal{H}$, σ_T and σ_T^s (the Cesaro mean and the Vallee-Poussin means) are defined respectively as the unique minimizers of the functions

$$\mathcal{G}_T(y) = \frac{1}{T} \int_0^T d^2(c(t), y) dt,$$

and

$$\mathcal{G}_T^s(y) = \frac{1}{T} \int_0^T d^2(c(t+s), y) dt.$$

A curve $c : [0, \infty) \rightarrow \mathcal{H}$ is called the Cesaro convergent or the mean convergent (resp. almost convergent) to $x \in X$, if σ_T (resp. σ_T^s) converges (resp. converges uniformly in s) to x as $T \rightarrow \infty$. It is easy to check that the results of Theorems 3.2 and 3.3 remain true for curves with similar proofs, but Theorem 3.5 for curves remains an open question.

Figure 1. Remark 3.1 and the open curve question, source PDF page 14.

The source does not spell out which continuous replacement for the discrete increment condition is intended. The result below gives the infinitesimal analogue and also settles the tempting unit-lag interpretation negatively.

2 Definitions and proof intuition

A nonnegative $g \in L^1_{\text{loc}}([0, \infty))$ is an *upper gradient along c* if

$$d(c(s), c(t)) \leq \int_s^t g(u) du \quad (0 \leq s \leq t).$$

Every locally absolutely continuous metric-space curve admits such a g ; one may take its metric derivative $g = |c'|$ almost everywhere.

Proof intuition

The key observation is a first-moment estimate for a CAT(0) barycenter: its distance from any comparison point is no greater than the average distance of the data from that point. Apply it with the comparison point $c(T)$. The upper-gradient inequality bounds the distance from $c(t)$ to $c(T)$ by the variation accumulated between t and T . Reversing the order of integration counts the speed at time s with weight s , producing exactly the continuous counterpart of the discrete Abel-summation term.

The unit-lag condition misses all motion inside each interval of length one. A periodic sine curve therefore has zero unit-lag increments while retaining nonvanishing oscillation.

3 The continuous Tauberian theorem

Theorem 1 (Curve analogue of Theorem 3.5). *Let $(\mathcal{H}, d)$ be a Hadamard space and let $c : [0, \infty) \rightarrow \mathcal{H}$ be locally absolutely continuous. Let $g \in L^1_{\text{loc}}([0, \infty))$ be a nonnegative upper gradient along c , and let σ_T be the Karcher mean of $c|_{[0, T]}$. Then for every $y \in \mathcal{H}$ and $T > 0$,*

$$d(c(T), y) \leq d(\sigma_T, y) + \frac{1}{T} \int_0^T t g(t) dt. \quad (1)$$

Consequently, if

$$\sigma_T \longrightarrow y \quad \text{and} \quad \frac{1}{T} \int_0^T t g(t) dt \longrightarrow 0, \quad (2)$$

then $c(T) \rightarrow y$. In particular, the conclusion holds if $t|c'(t)| \rightarrow 0$ in the essential sense.

Proof. We first prove the barycenter estimate

$$d(\sigma_T, z) \leq \frac{1}{T} \int_0^T d(c(t), z) dt \quad (z \in \mathcal{H}). \quad (3)$$

Put $b = \sigma_T$, $D = d(b, z)$, and

$$F(w) = \frac{1}{T} \int_0^T d^2(c(t), w) dt.$$

The CAT(0) strong-convexity inequality, applied on the geodesic from b to z , and the minimality of b imply the variance inequality

$$D^2 \leq F(z) - F(b). \quad (4)$$

For each t , the triangle inequality gives

$$d^2(c(t), b) \geq (D - d(c(t), z))^2,$$

and hence

$$d^2(c(t), z) - d^2(c(t), b) \leq 2D d(c(t), z) - D^2.$$

After integration and use of (4),

$$D^2 \leq F(z) - F(b) \leq \frac{2D}{T} \int_0^T d(c(t), z) dt - D^2.$$

If $D = 0$, (3) is immediate; otherwise division by $2D$ proves it.

Now set $z = c(T)$ in (3). The upper-gradient property and Tonelli's theorem give

$$\begin{aligned} d(\sigma_T, c(T)) &\leq \frac{1}{T} \int_0^T d(c(t), c(T)) dt \\ &\leq \frac{1}{T} \int_0^T \int_t^T g(s) ds dt \\ &= \frac{1}{T} \int_0^T s g(s) ds. \end{aligned}$$

The triangle inequality proves (1), and (2) proves the convergence assertion.

Finally, if $t|c'|(t) \rightarrow 0$ essentially, then its Cesaro averages tend to zero: split the integral at a fixed sufficiently large time, divide the fixed initial integral by T , and bound the remaining average by an arbitrary ε . Taking $g = |c'|$ completes the last assertion. $\square$

Remark 1 (A slightly weaker exact hypothesis). *The averaged condition in (2), rather than pointwise decay of $t|c'|(t)$, is all the proof requires. This is the direct continuous counterpart of the Cesaro expression that appears after Abel summation in the source proof of Theorem 3.5.*

4 The unit-lag interpretation is false

Proposition 1. *The hypotheses*

$$\sigma_T \rightarrow y, \quad T d(c(T), c(T-1)) \rightarrow 0$$

do not imply $c(T) \rightarrow y$, even for a bounded smooth curve in the Hadamard space $\mathbb{R}$.

Proof. Take

$$c(t) = \sin(2\pi t), \quad t \geq 0.$$

In $\mathbb{R}$, the Karcher mean for squared distance is the ordinary integral mean, so

$$\sigma_T = \frac{1}{T} \int_0^T \sin(2\pi t) dt = \frac{1 - \cos(2\pi T)}{2\pi T} \rightarrow 0.$$

Periodicity gives $c(T) = c(T-1)$ for every $T \geq 1$, so the unit-lag Tauberian expression vanishes identically. Nevertheless,

$$c(n + \tfrac{1}{4}) = 1, \quad c(n + \tfrac{3}{4}) = -1,$$

and the curve does not converge. $\square$

5 Verification, scope, and novelty

The argument is deterministic and has no computational dependency. It uses only CAT(0) strong convexity, minimality of the Karcher mean, the triangle inequality, the upper-gradient property, and Tonelli's theorem. The cases $d(\sigma_T, z) = 0$ and $d(\sigma_T, z) > 0$ are separated before division. No global boundedness assumption on c is used.

Theorem 1 answers the source's curve question for the natural metric-derivative analogue in every Hadamard space. Proposition 1 rules out the other immediate interpretation. The packet does not answer the source's separate questions about uniformly convex non-CAT(0) spaces or about $\triangle$ -almost convergence.

The run’s lightweight indexes were searched using the arXiv id, exact title, and the terms *almost convergence*, *Karcher mean*, *curve*, and *Tauberian*. A bounded web/arXiv search through 23 July 2026 used the exact title, the exact phrase “Theorem 3.5 for curves”, the phrase “Karcher mean curve Tauberian condition”, and the authors’ names. It found the source paper and later work using the same curve-mean framework, but no resolution of this question. This is not an exhaustive novelty certification.

6 Human-review recommendation

The packet should be reviewed by an analyst familiar with $CAT(0)$ barycenters and Tauberian theory. The mathematical proof is short; the main interpretive question is whether the source authors intended the metric-derivative or unit-lag translation. Both possibilities are resolved here.

References

- [1] H. Khatibzadeh and H. Pouladi, *Tauberian Conditions for Almost Convergence in a Geodesic Metric Space*, arXiv:1906.06988 (2019).